BASED ON HAWKINS LAB EXPERIMENT 715

THE RIFT TRACK

A DANGEROUS TRACKING CHALLENGE

LINE FOLLOWER ROBOT

RESONANCE 2026

LINE FOLLOWER ROBOT

Overview

- Team Specifications:
  - i. Each team must consist of 1 to 3 members.
  - ii. Only registered members are allowed to participate.
- Inspection & Registration:
  - i. Robots must be registered before the event.
  - ii. Each robot will be assigned an ID/number.
  - iii. Inspection must be completed before participation.
- Teams must report on time.
- Robots must pass inspection before the event.
- Any disputes must be reported to the coordinator immediately after the run.
- The coordinator's decisions are final.

Prize Pool : ₹10,000 (1st: ₹6,500 | 2nd: ₹3,500)
Entry : ₹200

Track Specifications:

- The track consists of a 20 mm wide black line on a white background.
- The track may include:
  1. Sharp turns.
  2. Intersections.
  3. Dotted path.
- Start , finish and checkpoints will be clearly marked.
- The track surface will be the same for testing and competition.
- The Qualifier and final round may / may not use the same track.

Robot Specifications:

- The robot must be fully autonomous (no manual control)
- Ready made bots or pre-assembled kits are not allowed .
- Maximum dimensions: 26 cm × 26 cm × 26 cm (+2 mm tolerance)
- Weight limit: 1 - 1.2 kg
- Maximum voltage: 24V
- The robot must not damage the track or surroundings.
- A start mechanism (switch) is mandatory.

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Competition Format:

- Each team will be given 2 minutes of testing time before each round.
  - ➔ No hardware changes allowed.
  - ➔ No extra tuning time for the first attempt.
- A round consists of 2 Attempts (runs).
- The team can give a second attempt back to back or after sometime.
- Sensor tuning time will be given prior to the second attempt .
- The best time out of 2 attempts will be considered.

Timing & Rules:

- Timing starts on the Referee's signal to start
- Robots must start within 5 seconds of the referee's signal.
- Timing stops at the finish line.
- Maximum time per run: 3 minutes.
- Exceeding time = failed attempt.
- Maximum 7 touches are allowed for each attempt with a penalty of +10 seconds.
- Bot must be placed at the previous checkpoint on every touch.
- Exceeding number of touches = failed attempt.

Disqualification Criteria:

- Leaving the track at any point
- Damaging the track
- Use of external control or communication
- Failure to comply with robot specifications

NOTE:

- The organizers/coordinators reserve the right to modify rules if necessary.
- Any changes in rules will be informed prior to the commencement of the competition.
- Any misconduct will lead to immediate disqualification.

Co-ordinators:

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